German University of Technology in Oman

Delivery Company System

Al-ghala Al Kalbani 24-0309
Dana Abu Quta 24-0768
Lama Al Maharibi 24-0747
Zainab Abdul Redha 24-0693
Maryah Al Mujaini 24-0542

System Overview

Purpose: Manage customer orders, assign drivers, and track deliveries

Feautures:

- Manage customer data
- Create and process orders
- Assign drivers dynamically
- Track delivery status
- Estimate delivery time (ETA)

Main classes

01
Customer

02
Order

03
Driver

04
Company

Coustomer Class

Role: Represents the person placing an order (stores customer information used in orders.)

Main data:
- Name
- Address
- phone number

Concepts used :
- (private data, public getters/setters)
- Constructors/Destructors for initialization
- Exception handling for invalid contact info

Relation: Composed inside Order (an order has-a customer).

Order class

Role: Handles all data related to a delivery order.

Attributes:

- orderID
- itemDescription
- status
- driver

Concepts demonstrated:

- Composition (contains a customer object)
- Operator overloading (<< for printing and == for comparing orders)
- Exception handling (invalid status or no driver)

Relation: Aggregates a Driver (pointer to driver)

Driver class

Responsibility: Represents delivery staff.

Key Data:

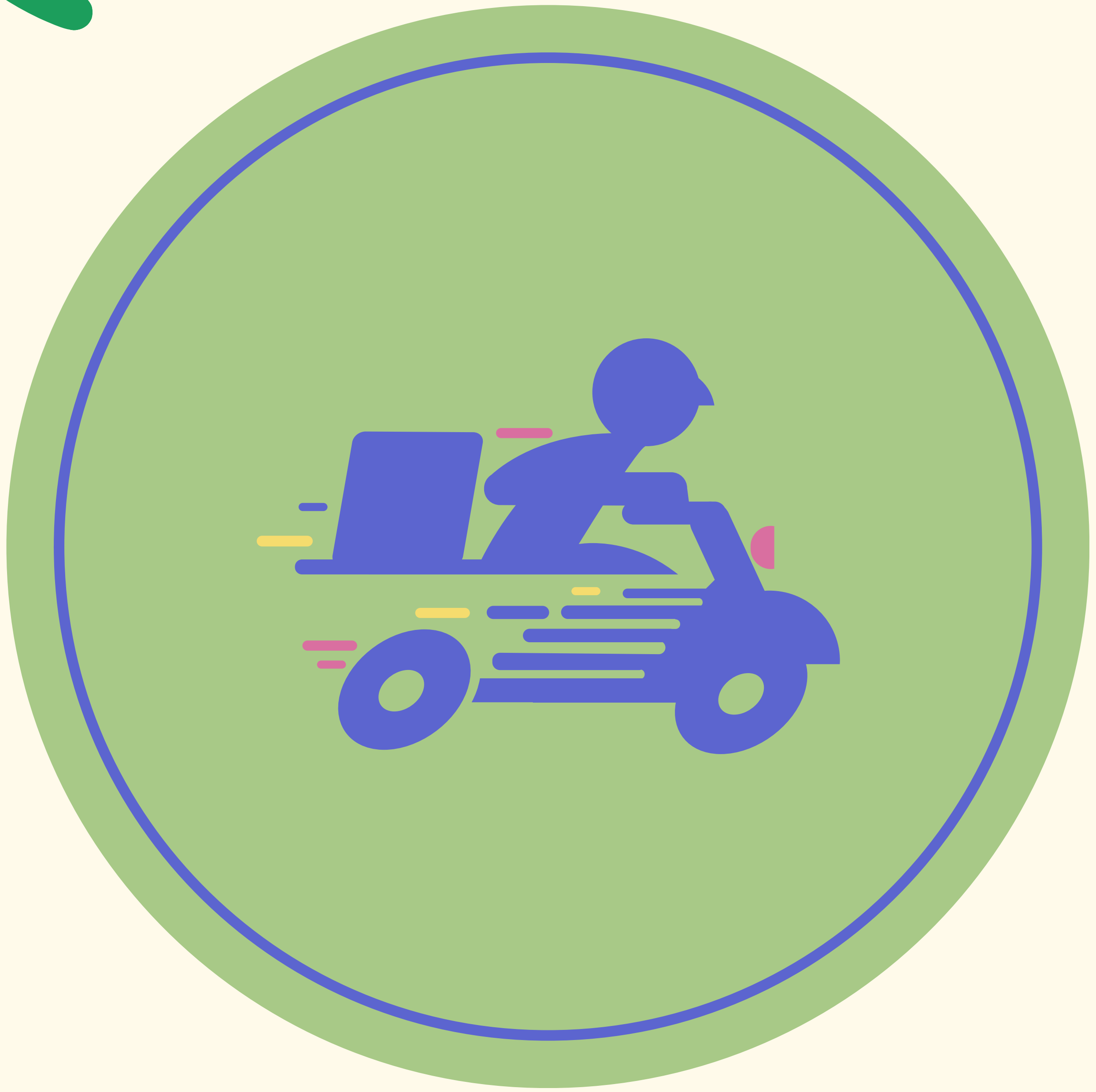
- Driver ID
- name
- vehicle type

Concepts Demonstrated:

- Inheritance (CarDriver, BikeDriver subclasses)

Relationship:

- Assigned to Orders
- Managed by Company



Company Class

- **Responsibility:** Central system managing orders, drivers, and budget (control the whole system)

- **Key Data:** Arrays of orders and drivers, budget objects.

- **Concepts Demonstrated:**
  - Composition (has a Budget)
  - (manages pointers to orders and drivers)
  - Exception handling (invalid IDs, unavailable drivers, budget overflow)
- **Relationship:** Connects all classes together.